PRESENTATION

Parking Lot Management System using Java

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BACKGROUND

With the rapid growth of vehicles due to urbanization, managing parking spaces has become a major challenge. Traditional manual parking systems are inefficient, time-consuming, and prone to errors in space allocation and fee calculation.

To overcome these issues, automated parking systems are needed. The Parking Lot Management System uses Java and Object-Oriented Programming (OOP) concepts to provide an efficient, accurate, and organized solution for managing parking operations.

PROJECT GOALS

1

To automate vehicle entry, exit, and parking space allocation

2

To apply Object-Oriented Programming (OOP) concepts in Java for efficient system design

3

To calculate parking fees accurately based on duration and vehicle type

INTRODUCTION

With the increasing number of vehicles in urban areas, managing parking spaces efficiently has become a significant challenge. Manual parking systems often lead to confusion, delays, and improper utilization of available space.

The Parking Lot Management System is designed to automate parking operations such as vehicle entry, parking allocation, fee calculation, and exit management. Developed using Java and Object-Oriented Programming (OOP) concepts, this system provides a simple, efficient, and reliable solution for managing parking spaces.

METHODOLOGY

The Parking Lot Management System is developed using a structured Object-Oriented approach in Java. The system is designed by identifying key components such as Vehicle, ParkingSpot, Ticket, and ParkingLot, and implementing them as classes.

The development follows these steps:

- Analyze system requirements and define functionalities
- Design class structure using OOP concepts like abstraction, inheritance, and encapsulation
- Implement core features such as vehicle entry, parking allocation, ticket generation, and fee calculation
- Use data structures like ArrayList and HashMap to store and manage data
- Test the system to ensure correct functionality and handle exceptions

This approach ensures the system is organized, efficient, and easy to maintain or extend in the future.

SYSTEM DESIGN

- Main Components
- Vehicle (Abstract Class): Base class for all vehicles
- Car & Bike: Derived classes with specific fee logic
- ParkingSpot: Represents each parking space
- Ticket: Stores vehicle entry details
- ParkingLot: Core class managing operations
- Main: User interface (menu-driven system)

ADVANTAGES

- Efficient space management
- Reduces manual errors
- Easy to use and understand
- Scalable and maintainable
- Demonstrates strong OOP concepts

LIMITATIONS

- No database integration (data not stored permanently)
- Console-based interface (no GUI)
- Limited to small-scale parking systems

FUTURE ENHANCEMENT

- Add graphical user interface (GUI)
- Integrate database (MySQL)
- Add multi-floor parking support
- Online booking/reservation system
- QR-based ticket system

CONCLUSION

The Parking Lot Management System successfully demonstrates the application of Object-Oriented Programming concepts in solving a real-world problem. The system efficiently manages parking operations, reduces manual effort, and ensures accurate fee calculation. With further enhancements, it can be developed into a full-scale commercial parking solution.

THANK YOU